A Short Analysis of the **HHL Algorithm**

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The setup and figures in this paper are taken from the *Introduction to Quantum Computation* course taught at EPFL during the 2024/2025 academic year. Any explanations related to (i) the case of a non-Hermitian matrix A, (ii) the propositions, and (iii) the measurements are solely my own.

Chapter 1

Introduction

The HHL (Harrow-Hasidim-Lloyd) algorithm is a quantum algorithm designed to solve systems of linear equations. Given a matrix $A \in \mathbb{C}^{2^n \times 2^n}$ and a vector $\mathbf{b} \in \mathbb{C}^{2^n}$, we want to find a vector $\mathbf{x} \in \mathbb{C}^{2^n}$ that verifies,

$$A\mathbf{x} = \mathbf{b}. \tag{1.1}$$

The HHL algorithm is exponentially faster than the classical methods when the matrix A is sparse. If s is the number of non-zero elements per row in A, then the quantum circuit can be constructed with $\mathcal{O}(ns^2)$ gates whereas the best classical algorithm runs in $\mathcal{O}(2^n s)$.

To solve the linear system $A\mathbf{x} = \mathbf{b}$ with a quantum circuit, we need to represent $\mathbf{b}$ and $\mathbf{x}$ by quantum states, and thus we need to scale them to unit length $\|\mathbf{b}\| = \|\mathbf{x}\| = 1$. Then, $\mathbf{b}$ may be represented by a state $|b\rangle$ using n qubit such that $|b\rangle = \sum_{i=0}^{2^n-1} b_i |i\rangle$. Here, the b_i are the components of $\mathbf{b}$ and $|i\rangle$ is a state in the computational basis corresponding to the binary decomposition of i (e.g. with 4 qubits, $|5\rangle = |0101\rangle$). The vector solution $\mathbf{x}$ can be then represented by the state $|x\rangle$ that verifies

$$|x\rangle = cA^{-1}|b\rangle, \tag{1.2}$$

where $c^{-1} = \|A^{-1}|b\rangle\|$ ensures that the state is normalized.

The HHL algorithm uses quantum phase estimation to encode the solution $\mathbf{x}$ into a quantum state. To do so, the algorithm requires that the matrix A be Hermitian. However, even if A is not Hermitian, we can still find a solution to the system by running the HHL algorithm on the larger system:

$$\widetilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}, \quad \widetilde{b} = \begin{pmatrix} b \\ 0 \end{pmatrix}. \tag{1.3}$$

Indeed, first remark that $\widetilde{A}$ is itself Hermitian since:

$$\widetilde{A}^\dagger = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}^\dagger = \begin{pmatrix} 0 & A^* \\ A^T & 0 \end{pmatrix}^T = \begin{pmatrix} 0 & (A^T)^T \\ (A^*)^T & 0 \end{pmatrix} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix} = \widetilde{A}. \tag{1.4}$$

Thus, we may run the HHL algorithm with $\widetilde{A}$. Now, we have to find a block vector $\widetilde{\mathbf{x}} = \begin{pmatrix} \alpha & \beta \end{pmatrix}^T$ with the blocks α, β such that $\widetilde{A}\mathbf{x} = \widetilde{b}$. Thus we require:

$$\widetilde{A}\widetilde{\mathbf{x}} = \widetilde{b} \Longrightarrow \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix} \Longrightarrow \begin{pmatrix} A\beta \\ A^\dagger\alpha \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix} \Longrightarrow \begin{cases} A\beta = b \\ A^\dagger\alpha = 0 \end{cases}. \tag{1.5}$$

Hence, we may choose $\widetilde{\mathbf{x}}$ to have the blocks $\alpha = 0$ and $\beta = \mathbf{x}$ so that $\widetilde{\mathbf{x}} = \begin{pmatrix} 0 & \mathbf{x} \end{pmatrix}^T$. This choice of $\widetilde{\mathbf{x}}$ will let us solve the HHL algorithm with a non-Hermitian matrix A for the problem $A\mathbf{x} = b$, where we choose the entries from $2^n + 1$ to 2^{n+1} in $\widetilde{\mathbf{x}}$ to find $\mathbf{x}$. Thus from now on, we will assume that A is a Hermitian matrix. Furthermore, as a result of the Spectral Theorem, there exists a set of orthogonal states $\{|u_i\rangle\}_{i=0}^{2^n-1}$ such that A may be written as

$$A = \sum_{i=0}^{2^n-1} \lambda_i |u_i\rangle \langle u_i| , \tag{1.6}$$

where the $\lambda_i \in \mathbb{R}$ are the eigenvalues of A and the $\{|u_i\rangle\}_{i=0}^{2^n-1}$ form an eigenbasis of A. The state $|b\rangle$ may also be written in the $\{|u_i\rangle\}_{i=0}^{2^n-1}$ basis and denoted by

$$|b\rangle = \sum_{i=0}^{2^n-1} \beta_i |u_i\rangle . \tag{1.7}$$

Let us now end this introduction by directly providing the solution $|x\rangle$ of the system. Since $A = \sum_{i=0}^{2^n-1} \lambda_i |u_i\rangle \langle u_i|$ we have that $A^{-1} = \sum_{i=0}^{2^n-1} \lambda_i^{-1} |u_i\rangle \langle u_i|$. As a result we obtain:

$$cA^{-1}|b\rangle = c \sum_{i=0}^{2^n-1} \sum_{j=0}^{2^n-1} \lambda_i^{-1} \beta_j |u_i\rangle \underbrace{\langle u_i | u_j \rangle}_{=\delta_{ij}} = c \sum_{i=0}^{2^n-1} \sum_{j=0}^{2^n-1} \frac{\beta_j}{\lambda_i} \delta_{ij} |u_i\rangle = c \sum_{i=0}^{2^n-1} \frac{\beta_i}{\lambda_i} |u_i\rangle =: |x\rangle . \tag{1.8}$$

Thus, $|x\rangle = c \sum_{i=0}^{2^n-1} \frac{\beta_i}{\lambda_i} |u_i\rangle$ is the sought solution of the system and where c is still the normalizing constant that will be determined later on.

Chapter 2

Analysis of the HHL Circuit

The HHL circuit is represented in the following figure:

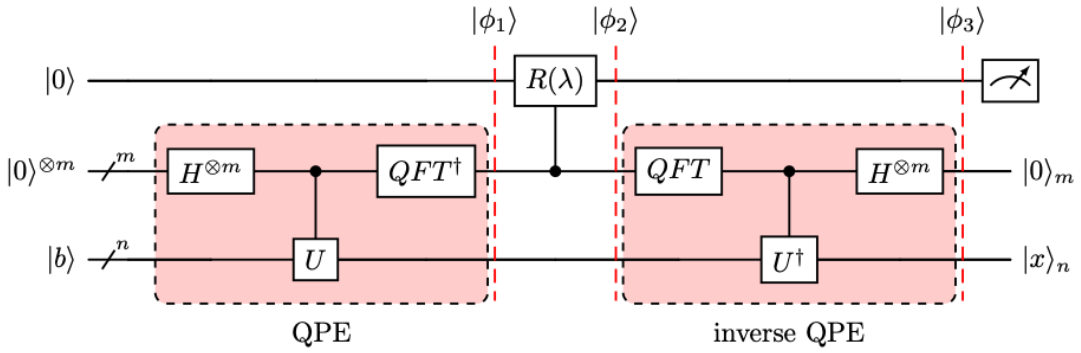


Figure 1: Quantum Circuit for HHL Algorithm

The top register is one ancilla qubit initialized to $|0\rangle$. The middle register is a memory register that stores the eigenvalues λ_i of A. More precisely, it will store the binary representation of λ_i, where the number of qubits m needed for this register will therefore depend on λ_i. This register is initialized to $|0\rangle^{\otimes m}$. Finally, the bottom register uses n qubits and is initialized with the state $|b\rangle$.
The circuit is composed of four steps: a quantum phase estimation (QPE), a controlled rotation, an inverse quantum phase and a measurement. We now detail the gates appearing in each part:

- **Quantum Phase Estimation**: This part of the circuit is detailed in the following figure.

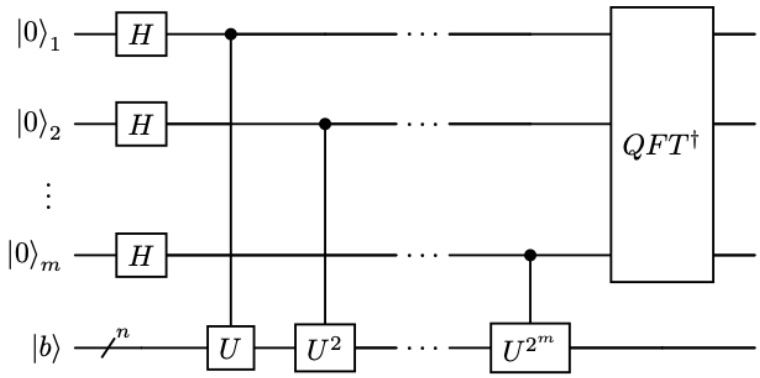


Figure 2: Detailed QPE

The circuit starts with a Hadamard gate on each qubit of the memory register. The unitary U is $U = \exp\left(2\pi i \frac{A}{2^m}\right)$. We then apply an inverse quantum Fourier transform on the memory register,

$$QFT^\dagger = \frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} e^{-2\pi i \frac{kj}{2^m}} |j\rangle . \tag{2.1}$$

- **Controlled rotation**: The gate R realizes the following transformation,

$$R(|0\rangle \otimes |\lambda\rangle) = \left(\sqrt{1 - \frac{1}{\lambda^2}}\,|0\rangle + \frac{1}{\lambda}\,|1\rangle\right) \otimes |\lambda\rangle . \tag{2.2}$$

- **Inverse QPE**: We apply the inverse gates of the QPE in reverse order to set back the memory register to $|0\rangle^{\otimes m}$. The memory register is no longer entangled with the output register.

- **Measurement**: The algorithm outputs $|x\rangle$ if the ancilla qubit is measured in state $|1\rangle$.

Let us now detail each step, and find the states after each gate. We will denote by $\{|\phi_i\rangle\}_{i=1}^3$ the output state after gate i. In what follows, we assume that the eigenvalues of A are positive integers, i.e. $\lambda_i \in \mathbb{N}^*$ and $\max_i(\lambda_i) < 2^m$. The circuit starts with a QPE that stores the eigenvalues in the memory register.

Proposition 2.0.1.

$$|\phi_1\rangle = |0\rangle \otimes \left(\sum_{j=0}^{2^n-1} \beta_j |\lambda_j\rangle \otimes |u_j\rangle\right). \tag{2.3}$$

Details. • As a first step, we find the eigenvectors and eigenvalues of the matrix $U = e^{2\pi i \frac{A}{2^m}}$. We then seek for vectors $|u\rangle$ and values λ such that $U|u\rangle = \lambda|u\rangle$. Using the definition of the exponential function as an infinite series we have that:

$$U|u_j\rangle = e^{2\pi i \frac{A}{2^m}}|u_j\rangle = \sum_{k\geq 0} \left(\frac{2\pi i}{2^m}\right)^k \frac{1}{k!} A^k |u_j\rangle . \tag{2.4}$$

Thus, we now seek for the eigenvectors and eigenvalues of A^k. For the case $k = 2$ we find the following:

$$A^2 = \left(\sum_{j=0}^{2^n-1} \lambda_j |u_j\rangle\langle u_j|\right)^2 = \sum_{j=0}^{2^n-1}\sum_{i=0}^{2^n-1} \lambda_j\lambda_i |u_i\rangle \underbrace{\langle u_i|u_j\rangle}_{=\delta_{ij}} \langle u_j| = \sum_{j=0}^{2^n-1} \lambda_j^2 |u_j\rangle\langle u_j| . \tag{2.5}$$

Thus, with a simple induction, it is not difficult to show that:

$$A^k = \sum_{j=0}^{2^n-1} \lambda_j^k |u_j\rangle\langle u_j| , \tag{2.6}$$

and therefore the eigenvalues of A^k are $\{\lambda_j^k\}_j$ and the associated eigenvectors are $\{|u_j\rangle\}_j$. We can now substitute the eigendecomposition of A^k to find the one of U:

$$U|u_j\rangle = e^{2\pi i \frac{A}{2^m}}|u_j\rangle = \sum_{k\geq 0} \left(\frac{2\pi i}{2^m}\right)^k \frac{1}{k!} A^k |u_j\rangle = \sum_{k\geq 0} \left(\frac{2\pi i}{2^m}\right)^k \frac{1}{k!} \lambda_j^k |u_j\rangle = \sum_{k\geq 0} \left(\frac{2\pi i \lambda_j}{2^m}\right)^k \frac{1}{k!} |u_j\rangle = e^{2\pi i \frac{\lambda_j}{2^m}}|u_j\rangle .$$

The eigenvectors of U are $\{|u_j\rangle\}_j$ with associated eigenvalues being $\left\{\exp\left(2\pi i \frac{\lambda_j}{2^m}\right)\right\}_j$.

• We now follow the flow of the QPE to start the calculation of $|\phi_1\rangle$. Assume that $|b\rangle$ is an eigenvector of U with eigenvalue $e^{2\pi i\varphi}$ where we consider the binary expansion of φ to be $\varphi = \sum_{j=1}^{m} \varphi_j 2^{-j}$. Thus, $U|b\rangle = e^{2\pi i\varphi}|b\rangle$ which implies that $U^{2^k}|b\rangle = e^{2\pi i\varphi 2^k}|b\rangle$. We apply in turns $H^{\otimes m}$ on the m qubits $|0\rangle$ and then apply a Control-U gate and find:

$$\begin{aligned}
|\phi_1\rangle &= \left(\mathbb{I}\otimes QFT^\dagger \otimes \mathbb{I}^{\otimes n}\right)\left(|0\rangle \otimes \bigotimes_{k=0}^{m-1} \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\varphi 2^k}|1\rangle\right) \otimes |b\rangle\right) \\
&= \left(\mathbb{I}\otimes QFT^\dagger \otimes \mathbb{I}^{\otimes n}\right)\left(|0\rangle \otimes \frac{1}{2^{\frac{m}{2}}} \sum_{z_1,\ldots,z_m\in\{0,1\}} \exp\left(2\pi i\varphi \sum_{k=1}^{m} z_k 2^{k-1}\right)|z_1 \ldots z_m\rangle \otimes |b\rangle\right) \\
&= \left(\mathbb{I}\otimes QFT^\dagger \otimes \mathbb{I}^{\otimes n}\right)\left(|0\rangle \otimes \frac{1}{2^{\frac{m}{2}}} \sum_{z=0}^{2^m-1} \exp\left(2\pi i\varphi z\right)|z\rangle \otimes |b\rangle\right).
\end{aligned}$$

Now, the goal is to make the QFT appear. For this, consider $\varphi = \sum_{j=1}^{m} \varphi_j 2^{-j} = \frac{1}{2^m}\sum_{j=1}^{m} \varphi_j 2^{m-j} = \frac{1}{2^m}\phi$, where $\varphi_1, \ldots, \varphi_m$ is the binary decomposition of $\phi \in \{0, \ldots, 2^{m-1}\}$. Thus, since QFT is unitary:

$$\begin{aligned}
|\phi_1\rangle &= \left(\mathbb{I}\otimes QFT^\dagger \otimes \mathbb{I}^{\otimes n}\right)\left(|0\rangle \otimes \frac{1}{2^{\frac{m}{2}}} \sum_{z=0}^{2^m-1} \exp\left(\frac{2\pi i\phi z}{2^m}\right)|z\rangle \otimes |b\rangle\right) \\
&= \left(\mathbb{I}\otimes QFT^\dagger \otimes \mathbb{I}^{\otimes n}\right)\left(|0\rangle \otimes QFT|\phi\rangle \otimes |b\rangle\right) \\
&= |0\rangle \otimes QFT^\dagger\left(QFT|\phi\rangle\right) \otimes |b\rangle = |0\rangle \otimes |\phi\rangle \otimes |b\rangle.
\end{aligned}$$

Note now that if $|b\rangle = |u_j\rangle$ we have that $U|b\rangle = U|u_j\rangle = e^{2\pi i\frac{\lambda_j}{2^m}}|u_j\rangle$. Hence, along with our original assumption that $U|b\rangle = e^{2\pi i\varphi}|b\rangle$, we require $2\pi i\varphi = 2\pi i\frac{\lambda_j}{2^m} \iff \varphi = \frac{\lambda_j}{2^m}$. And therefore for $|b\rangle = |u_j\rangle$ it holds that $\phi = 2^m\varphi = \lambda_j$. Thus since $|b\rangle = \sum_{i=0}^{2^n-1}\beta_i|u_i\rangle$ we end up with:

$$|\phi_1\rangle = |0\rangle \otimes |\phi\rangle \otimes |b\rangle = |0\rangle \otimes \left(\sum_{j=0}^{2^n-1} \beta_j |\lambda_j\rangle \otimes |u_j\rangle\right). \tag{2.7}$$

□

Proposition 2.0.2.

$$|\phi_2\rangle = \sum_{j=0}^{2^n-1} \beta_j \left(\sqrt{1-\frac{1}{\lambda_j^2}}|0\rangle + \frac{1}{\lambda_j}|1\rangle\right) \otimes |\lambda_j\rangle \otimes |u_j\rangle. \tag{2.8}$$

Details. We use the definition of the controlled rotation which is a gate taking a 2-qubit input and yields:

$$R\left(|0\rangle \otimes |\lambda\rangle\right) = \left(\sqrt{1-\frac{1}{\lambda^2}}|0\rangle + \frac{1}{\lambda}|1\rangle\right) \otimes |\lambda\rangle. \tag{2.9}$$

Thus, with an input of $|\phi_1\rangle$, we obtain:

$$\begin{aligned}
|\phi_2\rangle = \left(\text{Control-}R \otimes \mathbb{I}^{\otimes n}\right)|\phi_1\rangle &= \left(\text{Control-}R \otimes \mathbb{I}^{\otimes n}\right)\left[|0\rangle \otimes \left(\sum_{j=0}^{2^n-1} \beta_j |\lambda_j\rangle \otimes |u_j\rangle\right)\right] \\
&= \sum_{j=0}^{2^n-1} \beta_j R\left(|0\rangle \otimes |\lambda_j\rangle\right) \otimes |u_j\rangle = \sum_{j=0}^{2^n-1} \beta_j \left(\sqrt{1-\frac{1}{\lambda_j^2}}|0\rangle + \frac{1}{\lambda_j}|1\rangle\right) \otimes |\lambda_j\rangle \otimes |u_j\rangle.
\end{aligned}$$

□

Proposition 2.0.3.

$$|\phi_3\rangle = \sum_{j=0}^{2^n-1} \beta_j \left(\sqrt{1-\frac{1}{\lambda_j^2}}\,|0\rangle + \frac{1}{\lambda_j}\,|1\rangle \right) \otimes |0\rangle^{\otimes m} \otimes |u_j\rangle\,. \tag{2.10}$$

Details. For convenience, we set $|\Lambda_j\rangle := \left(\sqrt{1-\frac{1}{\lambda_j^2}}\,|0\rangle + \frac{1}{\lambda_j}\,|1\rangle \right)$ since this first part will not play a role in the inverse QPE. Thus, $|\phi_2\rangle$ becomes:

$$|\phi_2\rangle \equiv \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes |\lambda_j\rangle \otimes |u_j\rangle\,. \tag{2.11}$$

We will decompose the inverse QPE in intermediate steps, where $|\psi_1\rangle$ will be the state after the QFT, $|\psi_2\rangle$ the state after the controlled-$U^\dagger$ gate and finally $|\psi_3\rangle$ the output state of the QPE. We first start with $|\psi_1\rangle$:

$$\begin{aligned}
|\psi_1\rangle = \left(\mathbb{I}\otimes QFT\otimes \mathbb{I}^{\otimes n}\right)|\phi_2\rangle = \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes QFT\,|\lambda_j\rangle \otimes |u_j\rangle &= \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(\frac{1}{2^{\frac{m}{2}}} \sum_{u=0}^{2^m-1} e^{2\pi i \frac{\lambda_u u}{2^m}}\,|u\rangle \right) \otimes |u_j\rangle \\
&= \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(\frac{1}{2^{\frac{m}{2}}} \sum_{u=0}^{2^m-1} e^{2\pi i \varphi u}\,|u\rangle \right) \otimes |u_j\rangle\,.
\end{aligned}$$

We now move on to $|\psi_2\rangle$ where we apply a controlled-$U^\dagger$ gate on the $\{|u_j\rangle\}_j$, and also recall that $U^\dagger$ is the unitary gate defined by $U^\dagger = \exp\left(-2\pi i \frac{A^\dagger}{2^m}\right) = \exp\left(-2\pi i \frac{A}{2^m}\right)$:

$$\begin{aligned}
|\psi_2\rangle = \left(\mathbb{I}\otimes \text{Control-}U^\dagger\right)|\psi_1\rangle &= \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(\frac{1}{2^{\frac{m}{2}}} \sum_{u=0}^{2^m-1} e^{2\pi i \varphi u}\,|u\rangle \right) \otimes U^\dagger\,|u_j\rangle \\
&= \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(\frac{1}{2^{\frac{m}{2}}} \sum_{u=0}^{2^m-1} e^{2\pi i \varphi u}\,|u\rangle \right) \otimes e^{-2\pi i \frac{\lambda_j}{2^m}}\,|u_j\rangle \\
&= \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(\frac{1}{2^{\frac{m}{2}}} \sum_{u=0}^{2^m-1} e^{2\pi i u\left(\varphi - \frac{\lambda_j}{2^m}\right)}\,|u\rangle \right) \otimes |u_j\rangle \\
&= \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(\frac{1}{2^{\frac{m}{2}}} \sum_{u=0}^{2^m-1} |u\rangle \right) \otimes |u_j\rangle\,,
\end{aligned}$$

where we have used the fact that $\varphi = \frac{\lambda_j}{2^m}$. We finally apply $H^{\otimes m}$ on the m qubits where we remark that:

$$\frac{1}{2^{\frac{m}{2}}} \sum_{u=0}^{2^m-1} |u\rangle \equiv \left(H\,|0\rangle\right)^{\otimes m} \Longrightarrow |\psi_2\rangle = \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(H\,|0\rangle\right)^{\otimes m} \otimes |u_j\rangle\,. \tag{2.12}$$

For this last step, we apply $H^{\otimes m}$ and find:

$$\begin{aligned}
|\phi_3\rangle = \left(\mathbb{I}\otimes H^{\otimes m}\otimes \mathbb{I}^{\otimes n}\right) \left(\sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(H\,|0\rangle\right)^{\otimes m} \otimes |u_j\rangle \right) &= \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes H^{\otimes m}\left(H\,|0\rangle\right)^{\otimes m} \otimes |u_j\rangle \\
&= \sum_{j=0}^{2^n-1} \beta_j\,|\Lambda_j\rangle \otimes \left(H^2\,|0\rangle\right)^{\otimes m} \otimes |u_j\rangle
\end{aligned}$$

$$= \sum_{j=0}^{2^n-1} \beta_j \left|\Lambda_j\right\rangle \otimes \left(\mathbb{I}\left|0\right\rangle\right)^{\otimes m} \otimes \left|u_j\right\rangle$$
$$= \sum_{j=0}^{2^n-1} \beta_j \left|\Lambda_j\right\rangle \otimes \left|0\right\rangle^{\otimes m} \otimes \left|u_j\right\rangle .$$

Substituting the formula for $|\Lambda_j\rangle$ we obtain the correct output state for $|\phi_3\rangle$:

$$\left|\phi_3\right\rangle = \sum_{j=0}^{2^n-1} \beta_j \left(\sqrt{1-\frac{1}{\lambda_j^2}}\left|0\right\rangle + \frac{1}{\lambda_j}\left|1\right\rangle\right) \otimes \left|0\right\rangle^{\otimes m} \otimes \left|u_j\right\rangle . \tag{2.13}$$

□

Chapter 3

Measurement and Limitations

We finally arrive at the measurement part. Upon a measurement of the state $|1\rangle$, we see that the output state will be:

$$\sum_{j=0}^{2^n-1} \frac{\beta_j}{\lambda_j} |1\rangle \otimes |0\rangle^{\otimes m} \otimes |u_j\rangle = |1\rangle \otimes |0\rangle^{\otimes m} \otimes \left(\sum_{j=0}^{2^n-1} \frac{\beta_j}{\lambda_j} |u_j\rangle \right) = |1\rangle \otimes |0\rangle^{\otimes m} \otimes c^{-1} |x\rangle . \tag{3.1}$$

Thus, up to a normalization constant c which can be determined after the measurement, the output state in the last n qubits is exactly in the form of $|x\rangle$ encountered in (1.8). Hence, in the case where the result of the measurement is $|1\rangle$, the output of the circuit $|x\rangle$ is indeed a solution of the linear system $c^{-1}|x\rangle = A^{-1}|b\rangle$. We note that if we measure $|0\rangle$ instead, the computation may be repeated until we measure $|1\rangle$. We now find the explicit formula for the normalization constant c, where we impose that the solution $|x\rangle$ to the system is normalized such that $\langle x|x\rangle = 1$. Thus:

$$\langle x|x\rangle = c^2 \sum_{i=0}^{2^n-1} \sum_{j=0}^{2^n-1} \frac{\beta_i}{\lambda_i} \frac{\beta_j}{\lambda_j} \langle u_j|u_i\rangle = c^2 \sum_{i=0}^{2^n-1} \sum_{j=0}^{2^n-1} \frac{\beta_i}{\lambda_i} \frac{\beta_j}{\lambda_j} \delta_{ij} = c^2 \sum_{i=0}^{2^n-1} \left(\frac{\beta_i}{\lambda_i} \right)^2 = 1 \Longrightarrow c^{-1} = \sqrt{\sum_{i=0}^{2^n-1} \left(\frac{\beta_i}{\lambda_i} \right)^2}.$$

Now, the probability $\mathbb{P}(|1\rangle)$ of measuring state $|1\rangle$ is given by:

$$\mathbb{P}(|1\rangle) = \left| (|1\rangle\langle 1| \otimes \mathbb{I}) |\phi_3\rangle \right|^2 = \sum_{i=0}^{2^n-1} \left(\frac{\beta_i}{\lambda_i} \right)^2 . \tag{3.2}$$

Let us find a lower bound for this probability to get a quantity dependent on the number m of qubits in the memory register. Using the fact that $\max_i(\lambda_i) < 2^m$ it implies that $\max_i \left(\lambda_i^{-2}\right) > 2^{-2m} = 4^{-m}$. Thus, also using that the 2-norm $\|\cdot\|_2$ of $|b\rangle$ satisfies $\||b\rangle\|_2^2 = \sum_i |\beta_i|^2 = 1$, we find:

$$\mathbb{P}(|1\rangle) = \sum_{i=0}^{2^n-1} \left(\frac{\beta_i}{\lambda_i} \right)^2 = \sum_{i=0}^{2^n-1} \frac{|\beta_i|^2}{|\lambda_i|^2} > 4^{-m} \sum_{i=0}^{2^n-1} |\beta_i|^2 = 4^{-m} \||b\rangle\|_2^2 = 4^{-m}. \tag{3.3}$$

Hence, a lower bound for $\mathbb{P}(|1\rangle)$ using the fact that $\max_i(\lambda_i) < 2^m$ and $\||b\rangle\|_2^2 = 1$ is 4^{-m}.

Although the HHL algorithm offers exponential speed-up over classical methods, it has several limitations. First, it requires that the matrix A be sparse which is not a fact that we take into account in our analysis. Furthermore, we also need to have some a priori knowledge of the eigenvalues λ_i to have an exact and efficient algorithm. In some instances, the preparation of the quantum state $|b\rangle$ can be difficult and may require some additional resources. We remember also that this algorithm only gives a solution with some probability of error, where on a real quantum noisy device we will only have access to an approximate solution.